Rock-Paper-Scissors in Nature: Population Dynamics of Cyclic Competition in *Uta Stansburiana*

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Abstract

In 1996, Sinervo & Lively documented a striking natural phenomenon: the three male throat-color morphs of the side-blotched lizard (*Uta stansburiana*) cycle through dominance in sequence over a five-year period, with no single morph ever permanently prevailing; a biological manifestation of the rock-paper-scissors game [1]. This paper applies the symmetric three-species Gause–Lotka–Volterra competition model of May & Leonard [2] to characterize the full landscape of population dynamics possible under intransitive cyclic dominance, and to assess which dynamical regimes are most consistent with biological observations of *Uta stansburiana*. Through a complete equilibrium and stability analysis, we identify three qualitatively distinct parameter regions: stable coexistence of all three morphs, competitive exclusion by a single morph, and sustained cycling with no stable equilibrium. We show that the cycling regime itself admits two qualitatively different behaviors depending on the strength of competitive asymmetry: periodic limit cycles on a critical boundary, and growing-period cycling in the general case.

1. Introduction

In the inner Coast Range of California, male side-blotched lizards (*Uta stansburiana*) come in three throat-color morphs: orange, blue, and yellow. Each morph is associated with a distinct reproductive strategy. Orange-throated males are highly aggressive, defending large territories and out competing blue males for access to females. Blue-throated males are vigilant mate-guarders, successfully defending against the sneaker strategy of yellow males, who mimic females to gain entry into territories they could not win outright. Yellow males are sneakers, and out-compete orange males by infiltrating the under-guarded harems of orange males. The result is an intransitive cycle: orange beats blue, blue beats yellow, yellow beats orange. From 1990 to 1995, Sinervo & Lively [1] observed this cycle in natural populations, with each morph rising and falling in frequency in sequence. No morph ever permanently prevailed. The system, in their words, plays a dynamic "rock-paper-scissors" game. They claimed this cycle took approximately 12 years to complete. While Sinervo's work has been foundational to understanding the population dynamics of *Uta-Stansburiana*, there are clear limitations. His initial 1996 study took place on a 250-meter-long outcropping in Merced County, California, with meager annual observational sample sizes ranging from 62 to 127[1], and predicted a 12-year cycle period. A Subsequent studies in which Sinervo led in 2007 [3] covered a broader geographical span, but predicted a four-to-five year cycle. In 2010 a paper, monomorphic (one color) and dimorphic (two color) populations were discovered [4]. Given the limited elapsed time since the studies have begun, the disparity between claims of period length, and the existence of monomorphic and dimorphic populations, it is difficult to ascertain the long-term behavior of diverse *Uta-Stansburiana* populations. Side-blotched lizards are found across a massive area covering up to 2,500,000 square kilo-

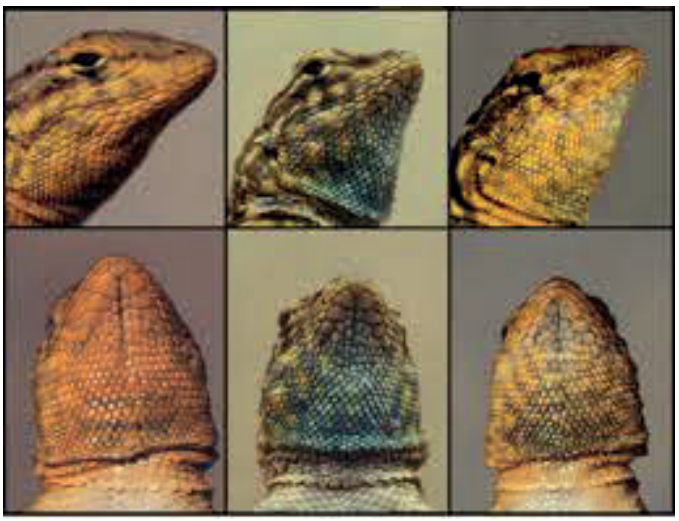


Figure 1: The three different color morphs of *Uta stansburiana*. From left to right: Orange (Aggressive), Blue (Vigilant Guarders), Yellow (Sneakers).

meters, ranging from California to Utah and south into Mexico, as outlined in Figure 2. Different sub-populations of the three morphs could exhibit varying levels of dominance over their competitors, yielding various population dynamics, including some that haven't been documented in the field. For that reason, this paper seeks to compre-

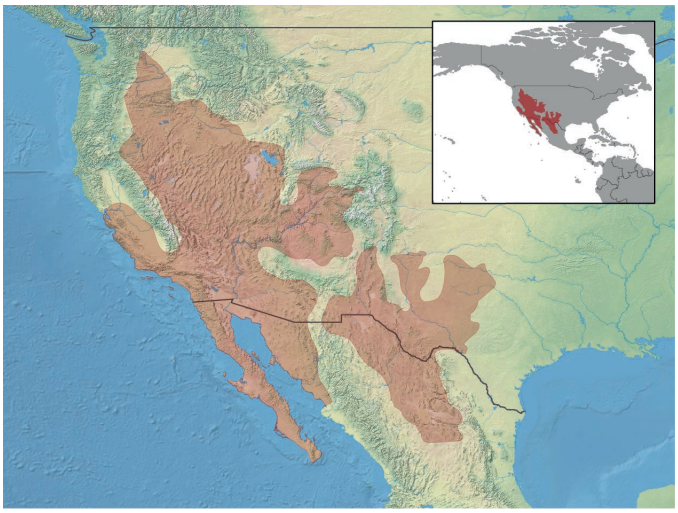


Figure 2: Geographic distribution of *Uta stansburiana* [5]

hensively model all the possible population dynamics that can be exhibited by an intransitive dominance relationship, i.e rock-paper-scissors.

The guiding question of this paper is: *What are the possible population dynamics for Uta stansburiana that can be predicted by an intransitive dominance model? Which regimes of the intransitive dominance model align closest with biological data? And what future behavior can this model predict?*

To do so, we follow the framework of May & Leonard's nonlinear analysis [2] of the three-species Gause-Lotka-Volterra [6, 7, 8] model. Their central result is that the competition parameters α and β, which encode the strength of suppression exerted by the loser and winner respectively in each pairwise interaction, partition the parameter plane into qualitatively distinct dynamical regimes. These different regimes are outlined in Figure 4. In one regime, all three species stably coexist. In another, competitive exclusion is inevitable, though which species wins depends on the initial population sizes. In a third, no equilibrium is stable and populations instead cycle indefinitely through near-dominance of each morph in turn. And within this cycling regime, both growing cycles and periodic limit cycles can arise depending on the parameters.

The remainder of the paper is organized as follows. Section 2 introduces the general GLV model and derives the symmetric two-parameter system through rescaling and symmetry assumptions, with the competition parameters α and β interpreted in the context of the lizard morphs. Section 3 finds all eight equilibrium points and the Jacobian matrix for the system. Section 4 conducts the stability analysis for the single species equilibria, outlining the competitive exclusion regime, with numerical simulations highlighting the basins of attraction. Section 5 conducts the stability analysis for the two-species equilibria, which results in an unstable and biologically infeasible regime. Section 6 conducts the stability analysis for the three-species equilibrium, outlining the regime in which the lizards can stably coexist. Section 7 analyzes the special boundary case that produces a family of periodic limit cycles. Section 8 treats the more general case that produces growing cycle-length estimates. Section 9 summarizes and discusses the paper's findings.

2. The Model

2.1. The General GLV System

The Gause–Lotka–Volterra equations [6, 7, 8] describing the dynamics of n competing populations consist of n first-order differential equations. For each species $i = 1, 2, \ldots, n$:

$$\frac{dN_i}{dt} = r_i N_i(t) \left[1 - \sum_{j=1}^{n} \alpha_{ij} N_j(t) \right]. \tag{1}$$

Where $N_i(t) \geq 0$ is the population of species i at time t; $r_i > 0$ is the intrinsic growth rate of species i, with units of time^{-1}. α_{ij} is the competition coefficient measuring the extent to which species j suppresses the growth of species i; α_{ii} is the self-competition (or self-regulation) coefficient of species i.

For $n = 3$ competing species, Eq (1) becomes:

$$\frac{dN_1}{dt} = r_1 N_1 \left[1 - \alpha_{11} N_1 - \alpha_{12} N_2 - \alpha_{13} N_3 \right], \tag{2}$$

$$\frac{dN_2}{dt} = r_2 N_2 \left[1 - \alpha_{21} N_1 - \alpha_{22} N_2 - \alpha_{23} N_3 \right], \tag{3}$$

$$\frac{dN_3}{dt} = r_3 N_3 \left[1 - \alpha_{31} N_1 - \alpha_{32} N_2 - \alpha_{33} N_3 \right]. \tag{4}$$

This system has 9 competition coefficients α_{ij} plus 3 growth rates r_i. We reduce this by setting $\tilde{N} = \alpha_{ii} N_i$, which normalizes the carrying capacities to 1, absorbing r into a rescaled time $\tilde{t} = rt$, and imposing cyclic symmetry by setting $\alpha_{12} = \alpha_{23} = \alpha_{31} = \alpha$ and $\beta_{12} = \beta_{23} = \beta_{31} = \beta$. The full rescaling and symmetry assumptions can be found in Appendix Section A. Substituting these assumptions into the rescaled system, and renaming the populations

in accordance with each lizard morph color, gives the final two-parameter model:

$$\frac{dN_O}{dt} = N_O[1 - N_O - \alpha N_B - \beta N_Y], \tag{5}$$
$$\frac{dN_B}{dt} = N_B[1 - \beta N_O - N_B - \alpha N_Y], \tag{6}$$
$$\frac{dN_Y}{dt} = N_Y[1 - \alpha N_O - \beta N_B - N_Y]. \tag{7}$$

Remark 1. *Typically, population dynamics literature refers to competing groups as "species." In our study of the side-blotched lizards, the three competing groups are all of the same species. Thus, for the rest of the paper, we will refer to the three competing groups as "morphs."*

2.2. *Uta Stansburiana* interpretation of α and β

The two parameters α and β encode the asymmetry of competitive suppression in the rock-paper-scissors cycle. In the pairwise interaction between any two morphs, one morph is the "winner" and one is the "loser"; that is, one suppresses the other more strongly than it is suppressed in return. The parameter β is the suppression coefficient exerted by the winner on the loser, and α is the suppression coefficient exerted by the loser back on the winner.

Applying these parameters to our lizard system: Orange males defend large territories and out compete blue males aggressively, suppressing the blue population with strength β, while blue males suppress orange males back with the weaker coefficient α. Blue males guard their mates vigilantly and suppress yellow "sneaker" males with strength β, while yellow males suppress blue males back with coefficient α. Yellow males exploit the under-guarded harems of orange males, suppressing orange with strength β, while orange males suppress yellow with coefficient α. The result is the intransitive cycle: orange β-suppresses blue, blue β-suppresses yellow, yellow β-suppresses orange, with each losing morph exerting weaker suppression α in return.

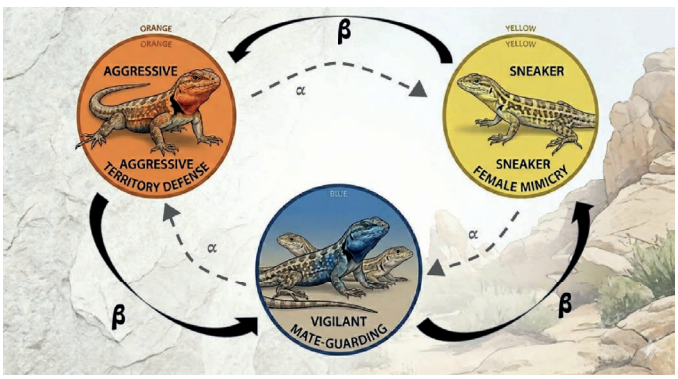


Figure 3: The rock-paper-scissors competitive structure of the three male throat-color morphs of the side-blotched lizard. *Generated using Google Gemini Nano Banana.*

3. Equilibrium Points and Jacobian Matrix

We now find the Equilibrium points of our system. Complete calculations can be found in the Appendix Section B.

Case 1: All moprhs absent. Gives the trivial equilibrium.

$$E_0 = (0, 0, 0).$$

Case 2: Exactly one morph present. Yields three single-morph equilibria of the form:

$$E_O = (1, 0, 0), \qquad E_B = (0, 1, 0), \qquad E_Y = (0, 0, 1).$$

Case 3: Exactly two morphs present. Gives three two-morph equilibria:

$$E_{OB} = \left(\frac{1-\alpha}{1-\alpha\beta}, \frac{1-\beta}{1-\alpha\beta}, 0\right),$$
$$E_{BY} = \left(0, \frac{1-\alpha}{1-\alpha\beta}, \frac{1-\beta}{1-\alpha\beta}\right), \tag{8}$$
$$E_{YO} = \left(\frac{1-\beta}{1-\alpha\beta}, 0, \frac{1-\alpha}{1-\alpha\beta}\right).$$

Case 4: All three morphs present. Gives the unique interior (coexistence) equilibrium

$$E^* = \frac{(1,\,1,\,1)}{1+\alpha+\beta}. \tag{9}$$

3.1. The Jacobian Matrix

To assess the stability of each equilibrium, we compute the Jacobian matrix of the system in Eqs (5)–(7). Taking partial derivatives of the right-hand side of each equation with respect to N_O, N_B, and N_Y, the Jacobian is

$$J(\mathbf{N}) = \begin{pmatrix} 1 - 2N_O - \alpha N_B - \beta N_Y & -\alpha N_O & -\beta N_O \\ -\beta N_B & 1 - \beta N_O - 2N_B - \alpha N_Y & -\alpha N_B \\ -\alpha N_Y & -\beta N_Y & 1 - \alpha N_O - \beta N_B - 2N_Y \end{pmatrix}. \tag{10}$$

In Sections 4-6, we evaluate Eq (10) at each equilibrium point to assess their stability. The results derived in Sections 4-8 partition the positive (α, β) parameter plane into three qualitatively distinct regions, visualized in Figure 4. In region (a), the interior equilibrium E^* is the unique stable point. In region (b), the three single-species equilibria E_O, E_B, E_Y are all simultaneously stable. In region (c), There are no asymptotically stable equilibrium points, yet the populations remain bounded. The detailed analysis of this behavior is the subject of Sections 7 and 8.

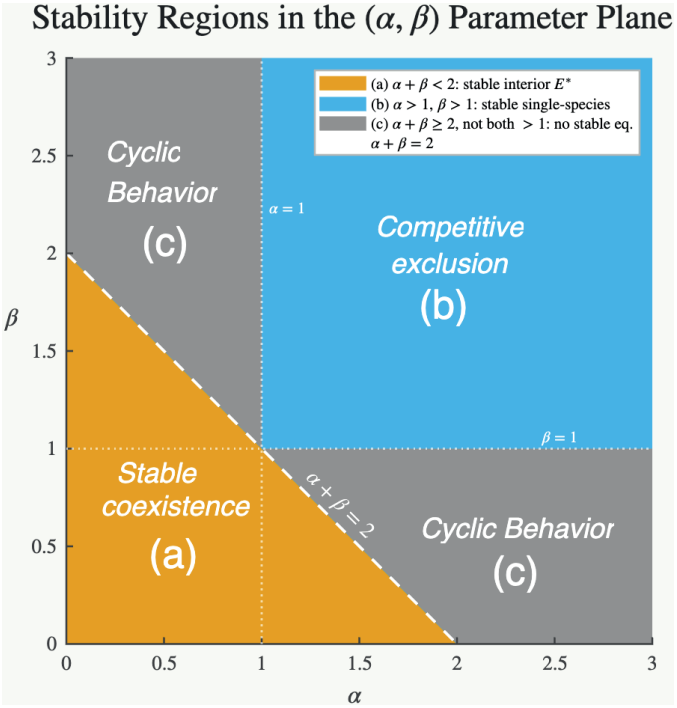


Figure 4: Stability regions of system (5)–(7) in the (α, β) parameter plane. **Orange (a):** all three species coexist. **Blue (b):** the system converges to competitive exclusion; one species survives. **Grey (c):** no equilibrium is stable; cyclic solutions arise. The dashed line marks the critical boundary $\alpha + \beta = 2$; dotted lines mark $\alpha = 1$ and $\beta = 1$.

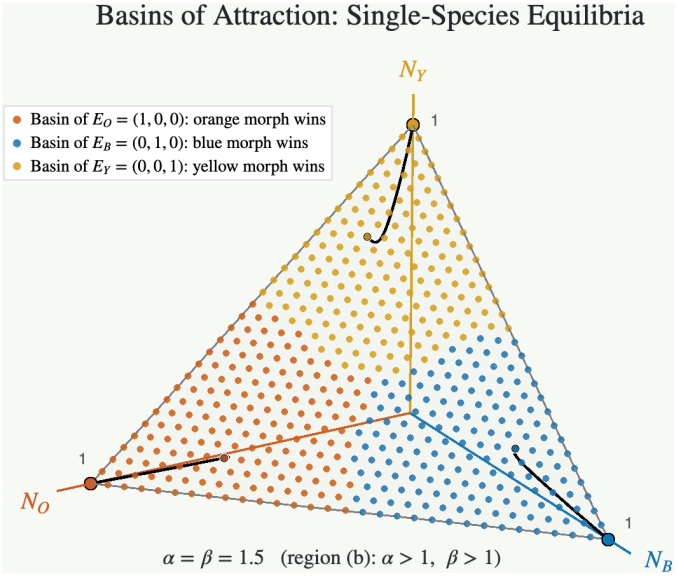


Figure 5: Basins of attraction for the three single-species equilibria of system Eqs (5)–(7) in region (b) ($\alpha > 1$ and $\beta > 1$). Each point on the simplex $N_O + N_B + N_Y = 1$ represents an initial condition $(N_O(0), N_B(0), N_Y(0))$, colored by the outcome of numerically integrating the system to $t = 500$: orange points converge to E_O, blue points converge to E_B, and yellow points converge to E_Y. The three black curves show representative trajectories, illustrating the path taken through population space before reaching the corresponding corner equilibrium.

4. Competitive Exclusion

We now assess the stability of the single-species equilibriums. We evaluate J at $E_O = (1, 0, 0)$:

$$J(E_O) = \begin{pmatrix} -1 & -\alpha & -\beta \\ 0 & 1-\beta & 0 \\ 0 & 0 & 1-\alpha \end{pmatrix}$$

This matrix is upper triangular, so its eigenvalues are read directly from the diagonal:

$$\lambda_1 = -1, \qquad \lambda_2 = 1 - \beta, \qquad \lambda_3 = 1 - \alpha.$$

We conclude that all three single-species equilibria are simultaneously stable if and only if

$$\boxed{\alpha > 1 \quad \text{and} \quad \beta > 1.}$$

A more detailed stability explanation can be found in Appendix Section C.1. When this condition holds, which of E_O, E_B, E_Y the system converges to depends on the initial conditions. The basins of attraction for each single species equilibrium are demonstrated in Figure 5. The basins show the set of all initial conditions from which the system's trajectory converges to each equilibrium as $t \to \infty$.

4.1. *Uta Stansburiana* Interpretation

Biologically, we can interpret this as a regime in which the pairwise dominance between the side-blotched lizard morphs is too extreme for coexistence. If the population begins in the yellow section (meaning yellow has a high initial population relative to the other morphs), the yellow morphs' sneaking and cuckoldry tactics are too effective for the orange morphs to defend. If we begin in the orange section, the orange morphs' aggressive territorial conquest is too effective for the blue morphs to guard against. Finally, if we begin in the blue section, the vigilant mate guarding of the blue morphs is too effective for the yellow morphs to infiltrate. Under these conditions, one morph dominates and the others are driven to extinction. Remarkably, this behavior is documented in nature. Corl et al. (2010) [4] surveyed 41 populations across the full range of *Uta stansburiana* and discovered 14 monomorphic populations. The single-species equilibria of our model offer a precise mathematical analogue to this phenomenon: when dominance parameters cross a critical threshold, one species dominates and the trajectory of the population is entirely determined by its initial conditions.

5. The Two Species Case

The three two-morph equilibria E_{OB}, E_{BY}, E_{YO} are never stable. The invasion eigenvalue at each reduces to $\lambda_3 = (1-\alpha)(1-\beta)$, which cannot be negative simultaneously with the conditions required for stability of any other equilibrium class. See Appendix C.2 for the full derivation. The instability of the two-species equilibira is simulated in Figure 6.

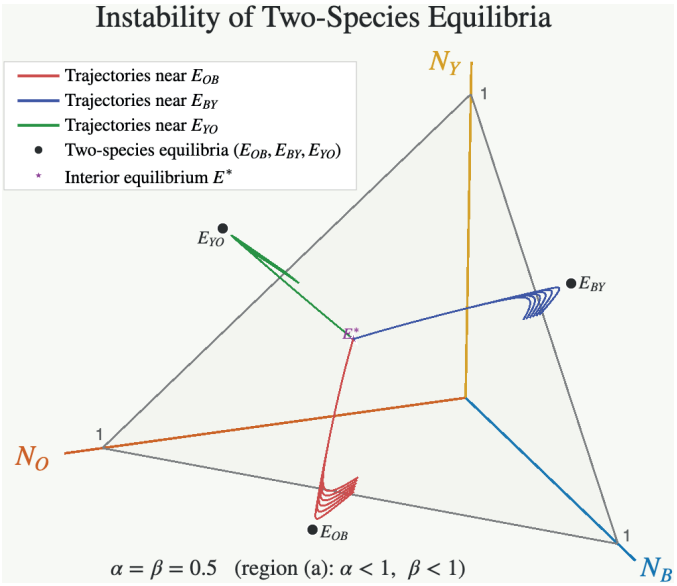


Figure 6: Numerical simulation illustrating the instability of the two-morph equilibria of system (5)–(7) in region (a) of Figure 4 ($\alpha + \beta < 2$). The three two-morph equilibria are each perturbed by introducing a small population of the absent morph in six increments ranging from $\varepsilon = 0.01$ to $\varepsilon = 0.06$. In each case the trajectory is repelled away from the two-morph equilibrium. All trajectories converge to the interior equilibrium E^*, consistent with the stability result found in Section 6. The increasing opacity of the curves corresponds to increasing perturbation size, showing that the repulsion occurs regardless of how small the initial invasion is.

5.1. *Uta Stansburiana* Interpretation

Each two-morph equilibrium represents a state in which one competitor has been driven to extinction, leaving the remaining two to interact in a one-sided dominance relationship. This asymmetry means that the dominant morph faces no check on its growth, and in the absence of the missing competitor, it will drive the subordinate morph to extinction as well. Crucially, these two-morph states are also unstable from the outside. The excluded morph is precisely the one with a competitive advantage over whichever morph currently dominates, meaning even a small number of returning individuals will grow rather than die out, destabilizing the two-morph state. Corl et al. [4] documented four dimorphic populations retaining only orange and blue morphs, which our model identifies as unstable. Our model predicts that these populations are either collapsing toward a single-morph outcome, or vulnerable to re-invasion by yellow should it reappear. The biology does not yet tell us which fate awaits these populations, but our model tells us it will not remain in that state.

6. Stable Coexistence

We now assess the stability of the interior three-morph equilibrium. The full analysis can be found in the Appendix Section C.3.

Recall that $E^* = (N^*, N^*, N^*)$ where $N^* = 1/(1+\alpha+\beta)$. Substituting into the Jaocbian Eq (10) gives us:

$$J(E^*) = -N^* \begin{pmatrix} 1 & \alpha & \beta \\ \beta & 1 & \alpha \\ \alpha & \beta & 1 \end{pmatrix} = -N^* \mathbf{A},$$

where $\mathbf{A}$ is a circulant competition matrix, a term coined Robert May's 1973 *Stability and Complexity in Model Ecosystems* textbook [9]. The circulant competition matrix eigenvalues are given by

$$\lambda_1 = 1 + \alpha + \beta,$$

$$\lambda_{2,3} = 1 - \frac{\alpha+\beta}{2} \pm i \left(\frac{\sqrt{3}}{2}\right)(\alpha - \beta).$$

Therefore, the interior equilibrium E^* is locally asymptotically stable if and only if

$$\boxed{\alpha + \beta < 2.}$$

The stability of the three-species equilibrium is numerically simulated in Figure 7.

6.1. *Uta Stansburiana* Interpretation

While there is no documented lizard population that exhibits stable coexistence at equal frequencies, the stability of E^* in the region $\alpha+\beta < 2$ is biologically sensible because under these conditions the suppression each morph exerts on others is collectively moderate. In the context of the side-blotched lizard system, this means that no morph is so effective at displacing its competitor that it can grow without bound before the third morph intervenes. The cyclic dominance structure remains intact, but the intensity of competition at each link is sufficiently tempered that the system reaches stable coexistence. Intuitively, when suppression is moderate, a morph that begins to increase in frequency is checked by the morph above it in the dominance hierarchy before it can dominate the population, and the same logic applies to each of the three. While there is no evidence of it, it is feasible that such populations exist.

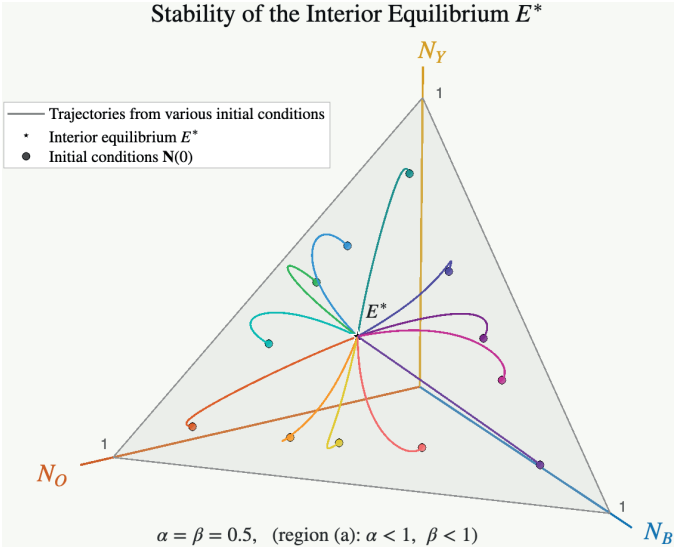


Figure 7: Numerical simulation confirming the local asymptotic stability of E^* of system (5)–(7) in region (a) of Figure 4 ($\alpha + \beta < 2$). Eighteen trajectories are integrated from initial conditions $\mathbf{N}(0)$, covering a wide range of starting configurations. Every trajectory converges to the unique interior equilibrium $E^* = (N^*, N^*, N^*)$ with $N^* = 1/(1 + \alpha + \beta) = 0.5$, regardless of where it begins.

7. Periodic Limit Cycles

We now analyze the special case $\alpha + \beta = 2$, which forms the boundary between region (a) and region (c) in Figure 4. The analysis proceeds in three steps: we show that the total lizard population $N_T = N_O + N_B + N_Y$ satisfies a logistic equation and converges to 1; we show that the product $P = N_O N_B N_Y$ converges to a constant C determined by the initial conditions; and we explain why these two constraints together force the asymptotic trajectory onto a closed curve. The complete derivation can be found in Appendix Section D.

We define the total population at time t:

$$N_T(t) \equiv N_O(t) + N_B(t) + N_Y(t). \tag{11}$$

For $\alpha + \beta = 2$, summing Eqs (5)–(7) gives:

$$\frac{dN_T}{dt} = N_T - N_T^2. \tag{12}$$

This is the standard logistic equation, which has the explicit general solution:

$$N_T(t) = \frac{N_T(0)}{N_T(0) + [1 - N_T(0)]\, e^{-t}}. \tag{13}$$

As $t \to \infty$, the exponential term $e^{-t} \to 0$, and so:

$$N_T(t) \to 1. \tag{14}$$

The total population converges to 1 regardless of the initial conditions. Geometrically, this means that the asymptotic trajectory lies on the plane $N_O + N_B + N_Y = 1$ in the 3-dimensional population space.

Next, we define the product of the three populations as

$$P(t) \equiv N_O(t)\, N_B(t)\, N_Y(t). \tag{15}$$

We derive an equation for P by working with the logarithmic form of the system:

$$\frac{d(\ln N_i)}{dt} = 1 - \sum_{j=1}^{3} \alpha_{ij} N_j. \tag{16}$$

Adding the three logarithmic equations gives

$$\frac{d(\ln P)}{dt} = 3(1 - N_T). \tag{17}$$

We now relate Eq (17) to the logistic Eq (12), which gives

$$\frac{d(\ln P)}{dt} = 3\, \frac{d(\ln N_T)}{dt}. \tag{18}$$

Integrating both sides with respect to t, we have

$$\ln P(t) = 3 \ln N_T(t) + \text{const}, \tag{19}$$

which gives

$$\frac{P(t)}{P(0)} = \left\{ \frac{N_T(t)}{N_T(0)} \right\}^3. \tag{20}$$

Taking $t \to \infty$ and using $N_T \to 1$ from (47):

$$P(t) \to P(0)\, [N_T(0)]^{-3} \equiv C. \tag{21}$$

The product $P(t) = N_O(t) N_B(t) N_Y(t)$ converges to a positive constant C that depends only on the initial conditions. The asymptotic trajectory therefore lies on the hyperboloid $N_O N_B N_Y = C$ in the 3-dimensional population space. Because $N_T \to 1$, every asymptotic trajectory is confined to the plane

$$\Pi : \quad N_O + N_B + N_Y = 1$$

in the three-dimensional population space. Because $P \to C$, every asymptotic trajectory is simultaneously confined to the hyperboloid

$$\mathcal{H}_C : \quad N_O N_B N_Y = C.$$

The asymptotic trajectory therefore lies on $\Pi \cap \mathcal{H}_C$, which is a closed curve. The trajectory is therefore a periodic limit cycle: . an isolated closed trajectory in the simplex to which nearby trajectories converge, corresponding to indefinite oscillations in population sizes. This behavior is numerically simulated in Figure 8, with the individual population curves outlined in Figure 9.

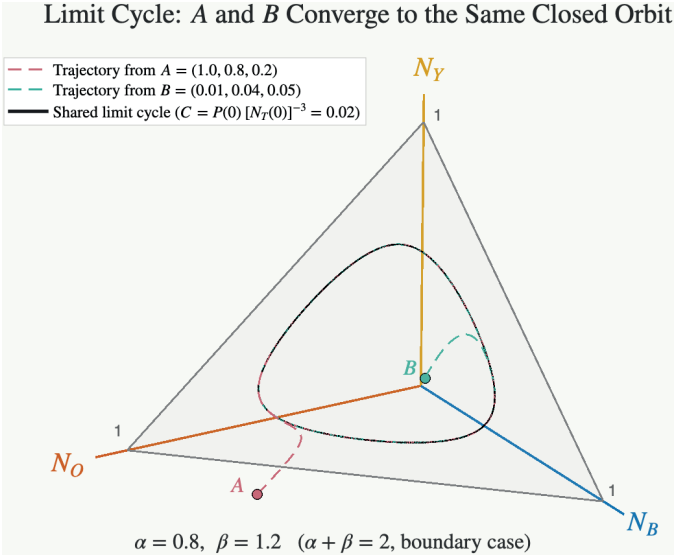


Figure 8: Numerical simulation of the boundary case $\alpha + \beta = 2$, illustrating the one-parameter family of periodic limit cycles established analytically in Section 7. Two trajectories are integrated from initial conditions $A = (1.0, 0.8, 0.2)$ and $B = (0.01, 0.04, 0.05)$, both chosen so that the $C = P(0)[N_T(0)]^{-3} = 0.02$ is identical. Despite starting in very different regions of population space, both trajectories converge to the same closed orbit (solid black curve). This confirms the analytic result of equation (56): the asymptotic trajectory is fully determined by C.

7.1. *Uta Stansburiana* Interpretation: Does This Match Observed Behavior?

The existence of periodic limit cycles in the $\alpha + \beta = 2$ regime of the intransitive dominance model offers perhaps the most direct correspondence with tri-morphic side-blotched lizard data. In his field studies of *Uta stansburiana* in California, Sinervo claimed that the relative frequencies of orange, blue, and yellow-throated males cycle with a period of approximately four to five years [3]. Biologically, the limit cycle reflects the inherent structure of cyclic dominance: orange-throated males spread through the population by out-competing blues, but as orange increases in frequency, yellow-throated sneakers gain the foothold they need to proliferate, which in turn creates conditions favorable for blue to recover, and the cycle begins again. Each morph's rise sows the conditions for its own eventual decline. While Sinervo's observation of a dominance cycle corresponds strongly to the periodic limit cycle regime, there is much room for uncertainty. Since he has begun his studies, there has been limited elapsed time to track many periods. In his initial 1996 paper, Sinervo claimed a cycle period of 12 years [1] and subsequently amended that claim in his 2007 paper [3] to "four-to-five" years, which is still not very precise. With the lack of precision in the period time, we present the possibility that the observed population cycles may not be periodic, but rather cycles that have growing periods, which we analyze in Section 8.

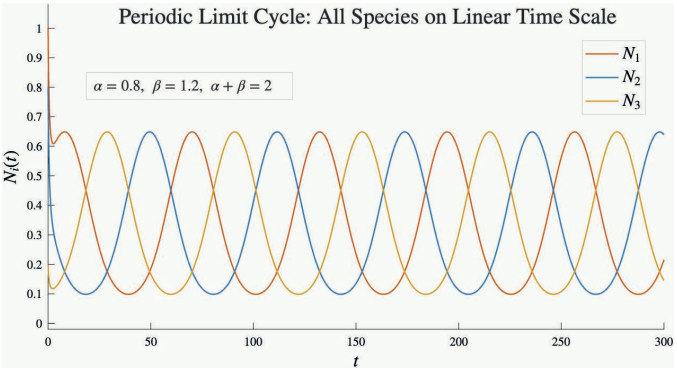


Figure 9: Population time series for all three morphs in the boundary case $\alpha + \beta = 2$ with $\alpha = 0.8$, $\beta = 1.2$, initialized at $\mathbf{N}(0) = (1.0, 0.8, 0.2)$. All three morph oscillate with constant amplitude and period, cycling through dominance in the order $N_1 \to N_2 \to N_3 \to N_1$ with a fixed phase offset of one-third of the period between each pair. The amplitude and period remain unchanged across the entire integration window $t \in [0, 300]$, confirming convergence to a stable limit cycle. This constant-period behavior distinguishes the boundary case from region (c) ($\alpha + \beta > 2$), where cycle periods grow without bound.

8. Growing Cycles

We now turn to the general case $\alpha + \beta > 2$, which corresponds to region (c) in Figure 4. In this regime, rather than converging to any fixed point, the populations cycle through near-dominance of each morph in turn, with the time required to complete each cycle growing proportionally to the total elapsed time. The complete analytical derivation can be found in the appendix E.

Following May & Leonard [2], we define the positive quantity

$$\gamma \equiv \alpha + \beta - 2 > 0, \tag{22}$$

which measures how far the parameters are from the boundary case. When $\gamma = 0$ we recover the $\alpha + \beta = 2$ boundary analyzed in Section 7; the larger γ is, the more strongly the system is in region (c). For definiteness we take $\beta > 1 > \alpha$ throughout this Section (so that $\beta > \alpha$).

8.1. The Total Population N_T Still Converges to 1

Adding equations (5)–(7) as before , but now without imposing $\alpha + \beta = 2$, yields

$$\frac{dN_T}{dt} = N_T(1 - N_T) + \gamma(N_O N_B + N_B N_Y + N_Y N_O), \tag{23}$$

where $\gamma = \alpha + \beta - 2$.

For large t, the product terms $N_O N_B$, $N_B N_Y$, $N_Y N_O$ are on average negligible compared to N_T. Intuitively, this is because for large t the system spends most of its time near the edges of the simplex, where at least one $N_i \approx 0$, making all three products small. This assumption is justified in the appendix of [2]. With this assumption, equation (23) reduces asymptotically to the same logistic equation as in the boundary case:

$$N_T \to 1 \quad \text{as } t \to \infty. \tag{24}$$

The total population still converges to 1, and the asymptotic trajectory still lies on the simplex $N_O + N_B + N_Y = 1$.

8.2. The Product $P(t)$ Decays Exponentially

The behavior of $P(t) = N_O N_B N_Y$ in the general case is fundamentally different from the boundary case. Adding the three logarithmic equations (49) as before, but now with $\alpha + \beta = 2 + \gamma$ gives

$$\frac{d(\ln P)}{dt} = 3(1 - N_T) - \gamma N_T.$$

Using $\frac{d(\ln N_T)}{dt} = 1 - N_T$ from (52) and the asymptotic result $N_T \to 1$, this becomes for large t:

$$\frac{d(\ln P)}{dt} \approx -\gamma. \tag{25}$$

Integrating from some reference time t_0:

$$\ln\left[\frac{P(t)}{P(t_0)}\right] \to -\gamma(t - t_0), \tag{26}$$

That is,

$$P(t) \to k\, e^{-\gamma t}, \tag{27}$$

where k is a positive constant. In contrast to the boundary case where $P(t) \to C$, here $P(t) \to 0$ exponentially. This means $N_O N_B N_Y \to 0$: at least one morph must approach zero. Since the system is constrained to the simplex $N_T = 1$ asymptotically, this means the trajectory is pushed toward the edges and corners of the simplex (the single-morph states). This is the key qualitative difference between the boundary case and the general case. When $\alpha + \beta = 2$, the product P converges to a positive constant C, and the trajectory settles on a periodic limit cycle in the interior of the simplex. When $\alpha + \beta > 2$, P decays to zero, and the trajectory is pushed ever closer to the corners, producing the growing-period cycling described below.

This growing cycle behavior is numerically simulated in the three-morph population space in Figure 10, and the individual population curves for that cycle are outlined in Figure 11. For Local Analysis Near a Single-Species State, see Appendix Section E.4. For a derivation estimating the time cycle, see Appendix Section E.5.

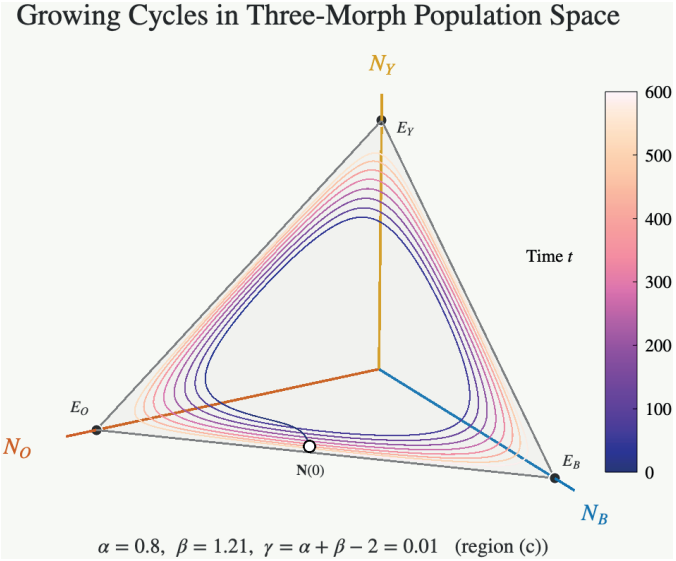


Figure 10: Growing cycles in three-morph in region (c) of the population space with $\alpha = 0.8$, $\beta = 1.21$ $\gamma = \alpha + \beta - 2 = 0.01$. The trajectory spirals outward from the initial condition $\mathbf{N}(0) = (0.6, 0.6, 0.1)$ toward the boundary of the simplex, visiting the neighborhoods of the single-morph equilibria in cyclic succession. Color encodes time from $t = 0$ (dark purple) to $t = 600$ (light pink), revealing that successive cycles grow progressively larger. Unlike the boundary case $\alpha + \beta = 2$, no stable limit cycle exists; the trajectory does not close.

8.3. *Uta Stansburiana* Interpretation: A Potential Alternative

The growing-cycle regime offers a biologically plausible alternative account of the dynamics observed in *Uta stansburiana*. The three morphs still cycle through dominance in sequence, but the time required to complete each successive cycle grows without bound. Ecologically, this means that each morph approaches near-extinction more closely with each pass, spending progressively longer intervals at very low frequency before the third morph intervenes to restore its advantage.

The empirical lizard data likely cannot distinguish this behavior from the periodic cycles of the boundary case, given the current state of the field record. With only a handful of cycles documented across roughly two decades of field study, and with sample sizes as low as 62 to 127 individuals in the original work, the data cannot reliably distinguish a fixed-period limit cycle from a slowly growing one. A growing cycle with small γ would appear nearly periodic over the timescales Sinervo observed, with successive periods differing by only a small fraction of the total cycle length.

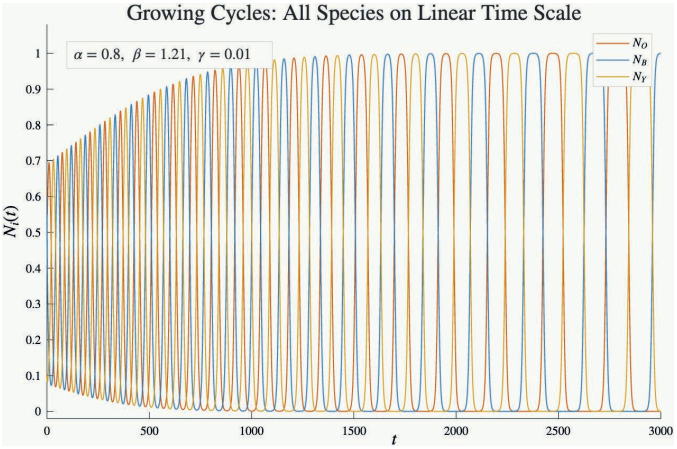


Figure 11: Population time series for all three morphs with $\alpha = 0.8$, $\beta = 1.21$, $\gamma = 0.01$. Each morph undergoes oscillations of growing amplitude and increasing period, with the three curves cycling through dominance $N_O \to N_B \to N_Y \to N_O$. As $t \to \infty$, the peaks approach carrying capacity while the troughs approach zero, reflecting the trajectory's asymptotic approach to the boundary of the simplex.

9. Discussion

This paper has applied the symmetric three-species Gause–Lotka–Volterra competition model of May & Leonard [2] to characterize the full landscape of possible population dynamics of *Uta Stansburiana*. We now synthesize these findings, discuss their biological interpretation, assess the weight of empirical evidence, and outline the limitations and open questions that remain.

Summary of Mathematical Results

In region (a) of Figure 4, the unique interior equilibrium $E^* = (N^*, N^*, N^*)$ with $N^* = 1/(1+\alpha+\beta)$ is locally asymptotically stable. All three morphs coexist at equal frequencies.

In region (b), all three single-morph equilibria are simultaneously locally asymptotically stable, and the system converges to competitive exclusion. Which morph ultimately wins depends entirely on the initial population composition. The two-morph equilibria E_{OB}, E_{BY}, E_{YO} are unstable throughout the parameter plane.

In region (c), defined by $\alpha + \beta > 2$, no equilibrium is asymptotically stable, yet the total population remains bounded and converges to the simplex $N_O + N_B + N_Y = 1$. The populations cycle indefinitely through near-dominance of each morph in turn. This cycling itself admits two qualitatively different behaviors. On the critical boundary $\alpha + \beta = 2$, the conserved quantity $P(t) = N_O N_B N_Y$ converges to a positive constant C determined by the initial conditions, and trajectories asymptotically approach a family of periodic limit cycles lying on the simplex. In the general case $\alpha + \beta > 2$, the product $P(t)$ decays exponentially, forcing trajectories toward the corners of the simplex and producing cycles of indefinitely growing period, with cycle time growing proportionally to the total elapsed time t.

Biological Implications for *Uta stansburiana*

The three parameter regions map onto biologically distinct scenarios for the side-blotched lizard system, and each deserves independent consideration.

Region (b), the competitive exclusion regime, was not been documented in any of Sinervo's field studies [1, 10, 3]. However, monomorphic populations were observed to exist in Corl's work [4]. *Uta stansburiana* occupies a vast range of approximately 2,500,000 square kilometers (Figure 2). Across this range, the ecological and evolutionary pressures on each morph's competitive strategy are not uniform. In isolated subpopulations where, for instance, the orange morph's territorial aggression confers a sufficiently large competitive advantage ($\beta > 1$) while simultaneously suppressing the yellow sneaker strategy less than in California populations ($\alpha > 1$), the dynamics drive the population to a single-morph outcome.

Region (a), stable coexistence, requires that the collective competitive suppression in the rock-paper-scissors cycle be moderate: $\alpha + \beta < 2$. When this holds, no morph can grow sufficiently unchecked before the morph with a competitive advantage over it rises to intervene. The result is a stable equilibrium at $E^* = (N^*, N^*, N^*)$. While no sustained periods of static equal-frequency coexistence have been reported in the lizard literature, region (a) is biologically feasible in tri-morphic populations that collectively have moderate dominance over each other.

The most directly relevant mathematical results for the tri-morphic lizard system are those of Sections 7 and 8, concerning the periodic limit cycle and growing cycle regimes of region (c).

Periodic Limit Cycles Versus Growing Cycles

A recurring theme in Sinervo's field record is the imprecision in estimated cycle period. His 1996 paper [1] predicted a cycle period of approximately 12 years; his 2007 follow-up [3] revised that estimate dramatically downward to four to five years. This correction is itself evidence of how difficult it is to pin down the true dynamics from a limited observational window.

The mathematical distinction between the boundary case ($\alpha+\beta = 2$, Section 7) and the general case ($\alpha+\beta > 2$, Section 8) is apparent in mathemetical theory but nearly invisible in practice for small $\gamma = \alpha+\beta-2$. In the boundary case, the period of oscillation is constant, and the conserved quantity $C = P(0)[N_T(0)]^{-3}$ fully determines the

asymptotic orbit. In the general case, the cycle time τ near each single-morph state grows proportionally to the total elapsed time. For small γ, the proportionality constant is small, and consecutive cycles differ in period by only a small fraction of the total cycle length. Over a two-decade observational window (roughly the span of Sinervo's work from 1990 onward) a system with $\gamma = 0.01$, for instance, would exhibit oscillations that appear nearly periodic, with successive periods differing by only a few percent. The data as currently collected cannot reliably distinguish this regime from the true boundary case. Whether the lizard parameters fall exactly on the $\alpha + \beta = 2$ boundary or slightly above it is an open empirical question that would require a substantially longer observational record, spanning many uninterrupted cycles and covering multiple geographically separated subpopulations, to resolve.

On the Symmetry Assumptions

The assumption of equal growth rates ($r_1 = r_2 = r_3$) is an idealization that warrants comment. In the actual lizard system, orange males are known to have higher testosterone levels and endurance than yellow or blue males [10], which plausibly affects their intrinsic growth rate. Similarly, territory size and mate-guarding intensity differ across morphs in ways that could break the cyclic symmetry of the α and β parameters.

Conclusion

Together, these results show that the three-species competition model (Eq (5)–(7)) captures a wide range of ecological outcomes for *Uta stansburiana*. Depending on the balance of competitive effects, the system may converge to stable coexistence, collapse to single-morph dominance, or enter sustained cycling.

While periodic cycling remains the most consistent with existing field observations, the mathematical boundary between stable coexistence and growing cycles is subtle enough that short-term datasets cannot definitively rule out alternative regimes in some subpopulations.

Overall, the model clarifies how small shifts in competitive interactions can move populations between qualitatively different dynamical behaviors. Determining which regimes are realized in nature, and how environmental and genetic variation modulates the parameters α and β, will require longer-term, geographically broad data.

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Full thesis on Middlebury Digital Commons

Evan Flaks



I first recall being excited by math in 5th grade when our school participated in the 24 Challenge – an arithmetic card game where players use four integers and basic operations $(+, -, \div, \times)$ exactly once to reach a total of 24, with the goal of being the first to solve the card. Although I was competitive, it wasn't the competition that drew me in. It was the first time math felt like something I actually wanted to engage with, rather than something assigned to me. Looking back, I think the 24 Challenge gave me my first real taste of mathematical problem-solving as opposed to mathematical procedure. There's a difference between executing an algorithm you've been taught and consciously searching for a solution. The satisfaction of uncovering an elusive solution is something I've pursued ever since through my coursework and ultimately my senior paper on nonlinear dynamical systems.

To those who claim "I am just not a math person," I'd argue that's impossible. Math is all around us and we interact with it everyday, whether we know it or not. When you split a bill, map out your day, adjust a recipe, or question the quality of a sale, you are doing math. When you are tapping your foot listening to music, you are doing math. Even the way your body calibrates balance and movement relies on ratios, angles, and predictions. You don't opt into math; it's embedded into how humans function. The question is not whether you are a math person, but whether you're looking to see the math within and around you.